# Multi-Agent Formation Control via Large Language Models and Reinforcement Learning

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***Abstract*—Formation control of multi-agent systems is a fundamental problem of swarm intelligence, aiming to enable a group of agents (such as drones, mobile robots, etc.) to form and maintain specific shapes through local perception and communication, while navigating toward target locations or accomplishing complex tasks. Over the past decades, extensive research efforts worldwide have advanced formation control from early theoretical studies and simulations to practical deployments in complex real-world environments. Correspondingly, methodologies have evolved from classical control methods to approaches that increasingly incorporate emerging technologies such as reinforcement learning (RL) and large language models (LLMs). A global research trend is the pursuit of higher levels of intelligence, improved robustness and safety, and practical applicability. Motivated by these developments, this study focuses on the trajectory planning problem in multi-agent formation control. By building upon state-of-the-art research, we propose a novel framework that integrates LLMs with RL. This hybrid approach aims to enhance decision-making and coordination capabilities in formation control tasks and to contribute to the development of more flexible and intelligent multi-agent systems.**

***Index Terms*—Multi-Agent Systems, Swarm Robotics, Formation Control, LLMs, Reinforcement Learning**

## I. Introduction

Multi-agent formation control is a fundamental problem in cooperative robotics, where a group of agents is required to arrange themselves into a desired geometric pattern while navigating through an environment. This capability is essential for a wide range of applications, including drone control [1] and warehouse management [2]. A critical yet often understudied subproblem within formation control is the target assignment problem, i.e., determining which agent should move to which position in the target formation. In conventional approaches, this assignment is either predefined by a human designer, computed based on proximity (e.g., the nearest-neighbor rule), or solved via centralized optimization algorithms such as the Hungarian method [3]. While effective in structured and static scenarios, these methods lack the flexibility to adapt to dynamic initial configurations, environmental constraints, or high-level shape descriptions provided in natural language.

Recent advances in Large Language Models (LLMs) have demonstrated their remarkable ability to understand spatial relationships, interpret semantic descriptions, and generate structured outputs [4], [5]. LLMs have been successfully applied to robot task planning, code generation, and even reward design in reinforcement learning [7]. However, their potential for automating formation target assignment remains largely unexplored. By leveraging LLMs, we can envision a system where a user simply describes the desired formation shape (e.g., "a triangle with 1-meter spacing" or "a line parallel to the x-axis") and provides the agents' initial positions, and the LLM automatically generates a feasible and efficient target assignment for each agent.

It is worth noting that many recent data-driven approaches to multi-agent coordination, particularly those based on multi-agent reinforcement learning (MARL), are built upon the Centralized Training with Decentralized Execution (CTDE) paradigm [8]. In CTDE, agents are trained with access to global information (e.g., joint observations, a central critic) to learn coordinated policies, which are then executed in a decentralized manner using only local perceptions. By this paradigm, we can effectively address the non-stationarity problem in decentralized learning and balance the need for global coordination with the practicality of distributed deployment.

This paper proposes a novel LLM-based framework for target assignment in multi-agent formation control. Our approach treats formation assignment as a semantic-to-spatial translation problem, where natural language descriptions of target shapes are mapped to concrete Cartesian coordinates for each agent, considering their initial distribution and optional constraints (e.g., limited vision field, local communication). The LLM is prompted with agent positions, environmental context, and formation specifications, and it outputs a target position list that minimizes the total travel distance, avoids collisions, and respects the integrity of the formation shape.

The main contributions of this work are as follows.

- A new LLM-based paradigm for formation target assignment that eliminates the need for manual design or rigid algorithmic assignment, enabling more intuitive and flexible formation specification.
- A structured prompting strategy that encodes geometric reasoning, agent states, and task constraints into LLM-readable format, improving assignment accuracy and spatial consistency.

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The rest of this paper is organized as follows. Section II reviews related work in formation control, target assignment algorithms, and LLMs in robotics. Section III formulates the LLM-based target assignment problem. Section IV details our methodology, including prompt design, LLM integration, and assignment refinement. Section V presents experimental results in diverse formation scenarios. Section VI discusses limitations and future directions, and Section VII concludes the paper.

## II. Related work

As early as 1987, Craig Reynolds proposed the Boids model [11], which established the idea of distributed, decentralized control and revealed the principle that simple local rules can give rise to complex global behavior. With the improvement of processing capabilities, various distributed approaches have emerged. For instance, Wang et al. proposed an innovative distributed method for local task exchange [12], which resolved potential deadlock problems in formation control and enhanced scalability. In recent years, with the continuous rise of large language models and reinforcement learning, researchers have also applied these powerful tools to formation control. In 2025, a multi-agent reinforcement learning method that leverages large language models to automatically generate and optimize reward functions online [9] is proposed, enabling automated policy learning for multi-agent formation control and obstacle avoidance tasks.

In summary, the field of formation control has undergone decades of development, evolving from classical control methods to a new stage of integration with emerging technologies. Recent research has consistently demonstrated a shared trend: a continuous push toward greater intelligence, the pursuit of more robust and safer methods, and an emphasis on real-world practicality.

In this context, this research aims to address the formation trajectory planning problem for multi-agent systems. Drawing on advanced achievements from both domestic and international research, we propose a method that integrates large language models with reinforcement learning, with the expectation of making a valuable contribution to the field.

## III. Preliminary & Methodology

### A. Reinforcement Learning

*1) Training Algorithm:* The Proximal Policy Optimization (PPO) algorithm is employed as the core training method for multi-agent reinforcement learning in this work. PPO is utilized within an Actor-Critic framework to optimize the policy network.

*a) Loss Function::* The loss function of PPO is defined as [9]:

$$\mathsf{loss}(\boldsymbol{\theta}^i) = \hat{\mathbb{E}}\Bigg[\min\Big(r_t(\boldsymbol{\theta}^i)\hat{A}_t^{\mathrm{GAE}(\gamma,\lambda)}, \\ \mathrm{clip}\big(r_t(\boldsymbol{\theta}^i), 1-\epsilon, 1+\epsilon\big)\hat{A}_t^{\mathrm{GAE}(\gamma,\lambda)}\Big)\Bigg],$$

where:

- $r_t(\boldsymbol{\theta}^i) = \dfrac{\pi(a_t, o_t; \boldsymbol{\theta}^i)}{\pi(a_t, o_t; \boldsymbol{\theta}^i_{\text{old}})}$ represents the probability ratio between the new and old policies.
- $\hat{A}_t^{\mathrm{GAE}(\gamma,\lambda)}$ is the Generalized Advantage Estimator, computed as:
$$\hat{A}_t^{\mathrm{GAE}(\gamma,\lambda)} = \sum_{l=0}^{\infty}(\gamma\lambda)^l \delta_{t+l}^V$$
with $\delta_t^V = r_t + \gamma V(s_{t-1}) - V(s_t)$ being the Temporal Difference (TD) error.
- $\gamma = 0.99$ is the discount factor.
- $\lambda = 0.95$ is the GAE discount factor.
- $\epsilon = 0.2$ is the clipping factor that restricts large deviations between the new and old policies.

*b) Training Framework::* The training is conducted under a Centralized Training with Decentralized Execution (CTDE) framework. During training, data from all agents are aggregated, and a shared reward is used, while each agent maintains its own individual policy network. During deployment, each agent makes decisions independently based on its own observations and policy network.

*c) Role in the Method::* PPO serves as the optimization algorithm that updates the policy network parameters $\theta_i$ based on the reward functions. The convergence of the loss function is used as an indicator that the agents' actions are being effectively guided by the current reward function. The algorithm enables stable and efficient policy learning for the formation control with collision avoidance task in dynamic multi-agent environments.

### B. Target Positions Generation

*1) LLM Selection:* The intersection of Large Language Models (LLMs) with various academic disciplines represents an emerging and highly active research frontier. Currently, LLMs have acquired substantial semantic knowledge and demonstrate formidable text generation capabilities. However, the quality and depth of content produced by different LLMs are constrained by their parameter scale and architectural design. For instance, smaller-scale models such as Qwen-7B, despite their efficiency, exhibit significantly inferior semantic comprehension and reasoning abilities compared to larger-scale models like DeepSeek and the GPT series (e.g., GPT-4). This performance gap is particularly critical in applications requiring high precision and contextual understanding. To ensure the effectiveness and reliability of the reinforcement learning model training process for our task of generating formation target coordinates, it is imperative to utilize a model with superior linguistic intelligence and instruction-following robustness. Consequently, we select a powerful GPT variant (e.g., GPT-4 or a comparable advanced model) as our core linguistic engine. This choice is motivated by its proven ability to interpret complex prompts accurately and generate coherent appropriate outputs, which is essential for producing valid and actionable coordinate specifications for multi-agent formation control.

*2) Prompt Engineering:* The prompts provided to the LLM should be detailed and specific. Below are the designed prompts [10]:

Listing 1. Prompt for the LLM-based coordinate generator.

```
You are an intelligent coordinate generator for swarm robot pattern formation. Your task is to generate a set of 2D coordinates that spatially form a target shape described in natural language.
The number of generated coordinates must exactly match the given number of agents, with no duplicates, and the coordinates must be within the specified arena size.

You will be given:
- Number of agents (N)
- Description of the target shape
- Arena size [width, height]

You must output only a list of N coordinates in the format [[x1, y1], [x2, y2], ..., [xN, yN]], where each x and y is an integer or float within the range [0, width] and [0, height] respectively. Do not include any other text or explanation.

Here are some examples:

Example 1:
Number of agents: 5
Target shape: Circle
Arena size: [100, 100]
Output: [[50, 90], [79, 65], [71, 29], [29, 29], [21, 65]]

Example 2:
Number of agents: 4
Target shape: Square
Arena size: [100, 100]
Output: [[25, 25], [25, 75], [75, 75], [75, 25]]

Example 3:
Number of agents: 3
Target shape: Triangle
Arena size: [100, 100]
Output: [[50, 90], [25, 25], [75, 25]]

Example 4:
Number of agents: 4
Target shape: Triangle
Arena size: [100, 100]
Output: [[50, 90], [25, 25], [75, 25], [50, 50]]
(You can put extra points on edges but maintain the overall shape.)

Now, for the following input:
Number of agents: {num_points}
Target shape: {shape_description}
Arena size: [{arena_size[0]}, {arena_size[1]}]
Output:
```

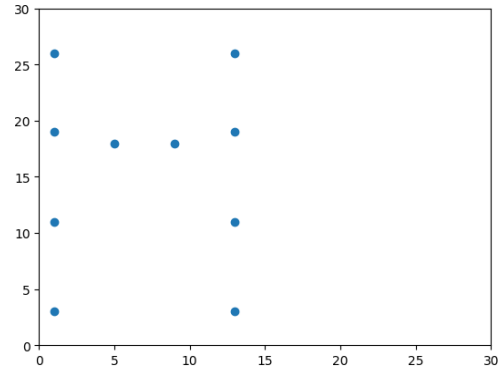


Fig. 1. Letter H in upper case

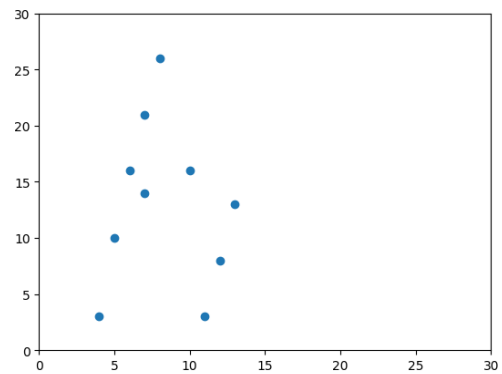


Fig. 2. Letter H in lower case

The "num_points", "shape_description", "arena_size[0], arena_size[1]" in the last three lines need to be replaced with environment parameters.

## IV. Experiments & Results

### A. Dataset

Our dataset, comprising 89 target shapes in total, was generated using an LLM. It encompasses a diverse set of categories: English letters (both uppercase and lowercase), digits, Greek letters, and fundamental geometric shapes.

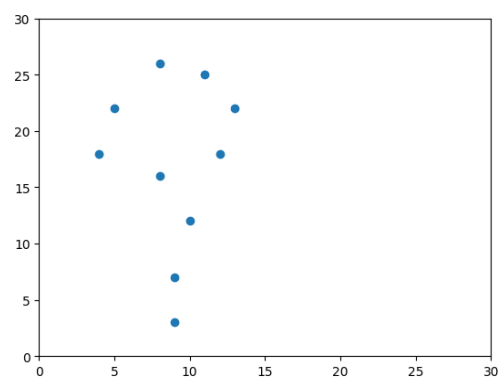


Fig. 3. Digit 9

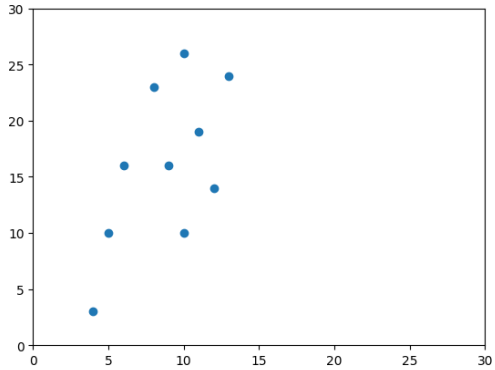

Fig. 4. Greek letter $\beta$

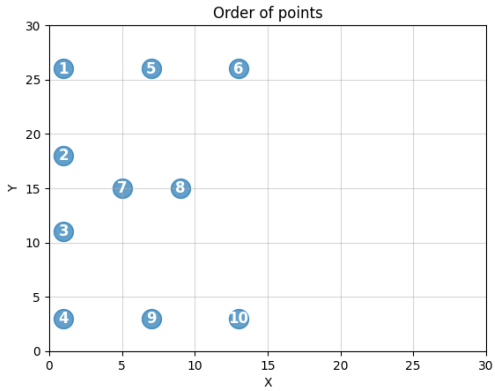


Fig. 5. Order 1 of letter E

## B. Order of Goal Points

A key finding is that the LLM does not produce a unique point sequence. As shown for the letter 'E', multiple valid orders (e.g., [[1, 26], [1, 18], [1, 11], [1, 3], [7, 26], [13, 26], [5, 15], [9, 15], [7, 3], [13, 3]] (shown in Fig.5.) and [[7, 26], [13, 26], [5, 15], [9, 15], [7, 3], [13, 3], [1, 26], [1, 18], [1, 11], [1, 3]] (shown in Fig.6.)) can represent the identical geometry. Crucially, this variation in input ordering leads to significant

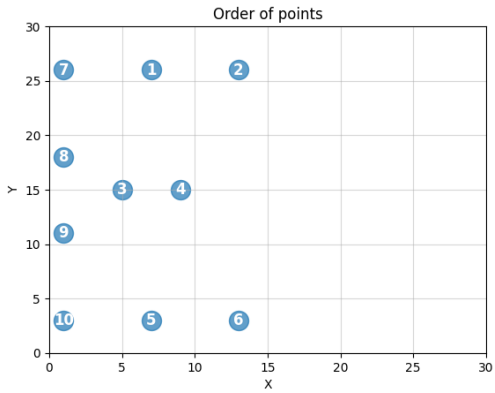


Fig. 6. Order 2 of letter E

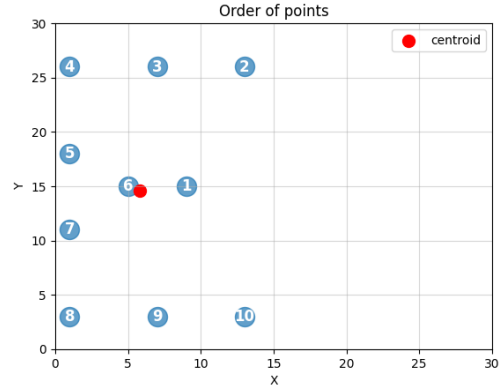


Fig. 7. Resorted order of letter E

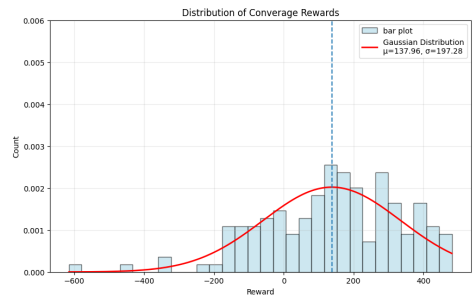


Fig. 8. Collected rewards of model without resorting and Gaussian distribution fitted to them

differences in model performance.

TABLE I
COMPARE BETWEEN 2 VERSION

| shape | S | U | D | G | O |
|---|---|---|---|---|---|
| rate1 | 0.76 | 0.74 | 0.24 | 0.49 | 0.68 |
| distance1 | 17.1830 | 17.8704 | 17.1120 | 17.4289 | 18.7810 |
| rate2 | 1.00 | 1.00 | 0.86 | 0.70 | 0.70 |
| distance2 | 16.9620 | 17.7163 | 15.4090 | 18.2227 | 16.3680 |

We first trained the model using the original sequence of points for each uppercase letter and tested it on held-out data. We found that for the letter B, the model achieved the best test performance. Consequently, we retrained the model on letter B using a reordered point sequence. As shown in Figs.

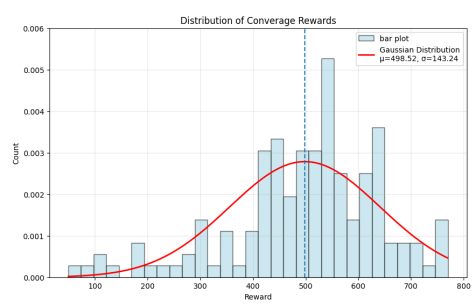


Fig. 9. Collected rewards of model with resorting and Gaussian distribution fitted to them

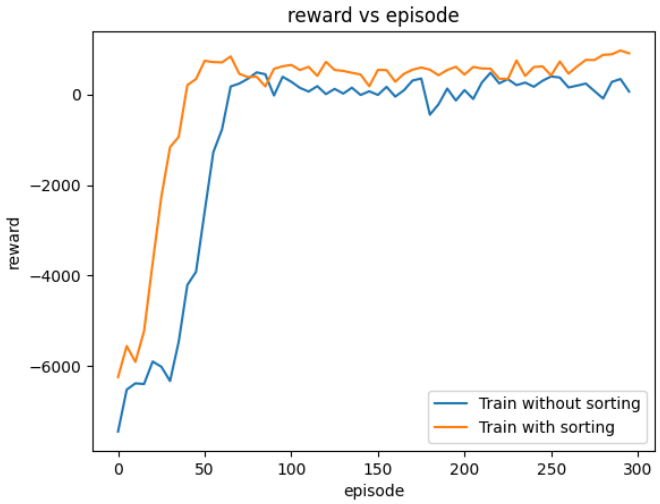


Fig. 10. Reward vs Episode

8 and 9, reordering the points leads to higher rewards during training. After both training processes converged, we collected 200 reward values. The model with point reordering achieved a significantly higher average reward (498.52) and a smaller standard deviation than the model without reordering (137.96). Table 1 compares the success rate (rate1 vs. rate2) and average distance (distance1 vs. distance2) of the two models on specific test letters. The results confirm that the model trained with the reordered point sequence performs better.

### C. Different Number of Shots

We trained the model using different numbers of samples, with the dataset split into training and testing sets. The model that achieved the best performance on the testing set was saved for comparison. As shown in Figs. 11 and 12, the converged reward improves slightly with more training samples. Table 2 further demonstrates that the success rate increases and the average moving distance decreases as the sample size grows.

TABLE II
PERFORMANCE OF MODELS WITH DIFFERENT NUMBER OF SHOTS

| #shot | 1 | 2 | 4 | 8 | 16 |
|---|---|---|---|---|---|
| success rate | 0.6427 | 0.7419 | 0.8031 | 0.8507 | 0.91 |
| distance | 21.433 | 19.732 | 18.003 | 17.299 | 16.783 |

## V. CONCLUSIONS & FUTURE WORK

In this paper, we study the formation control problem of multi-agent systems. Leveraging the semantic understanding and analysis capabilities of large language models (LLMs), we generate target formation coordinates for formation control tasks, providing a clear objective for the control mission. Subsequently, reinforcement learning is employed to train a model that controls multiple agents to move towards their target positions, thereby achieving multi-agent formation control.

During the model training and deployment process, we observed a significant lack of generalization in the model,

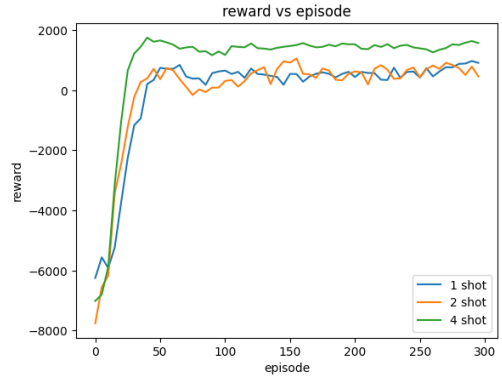


Fig. 11. Reward curves of 1 shot, 2 shots and 4 shots

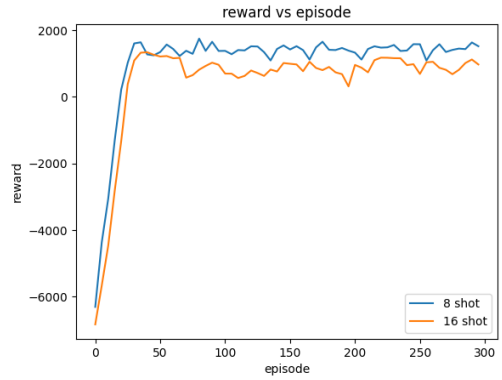


Fig. 12. Reward curves of 8 shots and 16 shots

which becomes especially pronounced when the number of trained agents increases. Furthermore, as the number of agents grows, the model's training performance declines sharply, with progressively worse convergence, ultimately affecting the success rate of formation control.

This research on formation control is a preliminary exploration. Based on the current findings, we plan to further improve the model's generalization and training capabilities in the future:

### A. Improve Model Generalization

In the next research phase, we will optimize the strategy for sharing models among agents to enable multiple agents to utilize the same action network.

### B. Optimize the Reward Function

Regarding the optimization of the reward function, we plan to enhance model training effectiveness and convergence capability by modifying communication strategies among agents and refining optimization objectives.

### C. Quantify Model Evaluation

Additionally, we also plan to add quantitative metrics such as success rate, the total distance moved by each agent during formation control, and other related content to further visualize the entire task.

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